

Hands-On: Understanding Component Lifecycle in React

Requirements

- Basic knowledge of React
- Text editor and web browser
- Node.js and npm installed on your computer

Instructions

1. Create a new React project using `create-react-app`.

```
npx create-react-app component-lifecycle-lab  
cd component-lifecycle-lab
```

2. Open the project folder in your text editor.

3. In the `src` folder, create a new file named `LifecycleComponent.js`.

```
import React, { Component } from "react";  
  
class LifecycleComponent extends Component {  
  constructor(props) {  
    super(props);  
    console.log("Constructor called");  
    this.state = { name: "React" };  
  }  
  
  componentDidMount() {  
    console.log("ComponentDidMount called");  
  }  
  
  componentDidUpdate(prevProps, prevState) {  
    console.log("ComponentDidUpdate called");  
  }  
  
  componentWillUnmount() {  
    console.log("ComponentWillUnmount called");  
  }  
  
  render() {  
    console.log("Render called");  
    return <div>Hello, {this.state.name}!</div>;  
  }  
}
```

```
export default LifecycleComponent;
```

4. In the `App.js` file, import the `LifecycleComponent` and render it in the `render()` method.

```
import React from "react";
import LifecycleComponent from "./LifecycleComponent";
```

```
function App() {
  return (
    <div className="App">
      <LifecycleComponent />
    </div>
  );
}
export default App;
```

5. Start the development server.

```
npm start
```

6. Open your browser and go to `http://localhost:3000/`. Open the developer tools and go to the console tab.
7. You should see the following output in the console:

```
Constructor called
Render called
ComponentDidMount called
```

This means that the constructor is called when the component is first created, the `render()` method is called to create the initial HTML, and the `componentDidMount()` method is called when the component is added to the DOM.

8. Now, change the state of the component by adding the following code at the end of the `render()` method:

```
<button onClick={() => this.setState({ name: "World" })}>
  Change Name
</button>
```

9. Save the file and go back to the browser. Click on the "Change Name" button. You should see the following output in the console:

```
Render called
ComponentDidUpdate called
```

This means that the `render()` method is called again with the updated state, and the `componentDidUpdate()` method is called after the component is updated in the DOM.

10. Finally, remove the `LifecycleComponent` from the `App` component and save the file. You should see the following output in the console:

```
ComponentWillUnmount called
```

This means that the `componentWillUnmount()` method is called when the component is removed from the DOM.

Congratulations! You have successfully completed the hands-on lab for understanding the component lifecycle in React.